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1 Demonstration

1.1 Notation

If Q satisfies all the conditions of a monoid except sharpness, we write $Q^\sharp$ for the quotient of Q by its subgroup of invertible elements. A monoid homomoprhism $f : P \to Q$ is called *sharp* if 0 is the unique $x \in P$ such that $f(x) = 0$. When $f : P \to Q$ is a monoid homomorhism, the quotient of P by all positive elements that project to 0 in Q is called the sharpening of f. A *valuation* of a monoid is a homomorphism to a valuative monoid.

We write $\operatorname{Rhiz} P$ for the site $\operatorname{Sh}(A_P)$ and $\operatorname{Spec} P$ for the site $\operatorname{Sh}(B_P)$.

1.2 Restatement of the theorem

We answer the question in the affirmative :

Theorem 1.2.1. *Let P be a monoid. For all presheaves of abelian groups F on $\operatorname{Spec} P$,*

$$\begin{aligned} &F(P) \to H^0(\operatorname{Rhiz} P, \pi^{-1}F) && \textit{is an isomorphism, and} \\ &H^i(\operatorname{Rhiz} P, \pi^{-1}F) = 0 && \textit{for all } i > 0. \end{aligned}$$

Applying this with P replaced by $P[-x]^\sharp$ gives required equivalence over every distinguished open.

1.3 Preliminaries

Lemma 1.3.1. *Infinitesimal homomorphisms are sharp.*

Proof. Suppose $f : M \to N$ is infinitesimal, that $y \in M$, and that $f(y) = 0$. Then $-y \in M^{\mathrm{gp}}$ and $f(-y) = 0 \in N$ and $-y \prec y$ and $f(y) \prec f(-y)$ so $-y \in M$. Since M is sharp, $y = 0$. □

Lemma 1.3.2. *Let $P \to Q \to V$ be monoid homomorphisms. Assume that $P \to Q$ is of finite presentation and let $\overline{P}$ and $\overline{Q}$ be the sharpenings of $P \to V$ and $Q \to V$, respectively. Then $\overline{P} \to \overline{Q}$ is also of finite presentation.*

Proof. We replace P with $\overline{P}$ and Q with the pushout of $P \to Q$ along $P \to \overline{P}$. Then we can assume without loss of generality that $P \to V$ is sharp.

Let $f_1, \ldots, f_n$ generate Q relative to P. Suppose that $h = g + \sum a_i f_i$ projects to 0 in V with g in the image of P and $a_i \in \mathbf{N}$. Then $0 \le g \le h$, so if h projects to 0 in V, so does g. Therefore $g = 0$ since $P \to V$ is sharp. Likewise, $\sum a_i f_i$ projects to 0 in V if and only if f_i projects to 0 for every nonzero a_i. Therefore the sharp localization $\overline{Q}$ of Q is determined by the subset of $\{f_1, \ldots, f_n\}$ that projects to 0. This subset is necessarily finite, so $Q \to \overline{Q}$ is finitely presented. It follows that $P \to \overline{Q}$ is a composition of finitely presented homomorphisms, hence is of finite presentation. □

Lemma 1.3.3. *Let $P \to Q$ be a monoid homomorphism of finite presentation, and let $\overline{P}$ be its sharpening. Then $P \to \overline{P}$ is also of finite presentation.*

Proof. We can form Q relative to P by embedding P^{gp} inside an abelian group A, then enlarging P as a semigroup by adjoining finitely many elements $f_1, \ldots, f_n$ of A, and finally saturating and sharpening the result. (Aribtrary relations $x = 0$ can be imposed by adjoining x and $-x$ and then sharpening.) Let B be the saturated subgroup of A spanned by $f_1, \ldots, f_n$. Within B, the elements $f_1, \ldots, f_n$ span a rational polyhedral cone R, of finite dimension. The intersection $-R \cap P$ spans a vector space of finite dimension in $\mathbf{Q} \otimes P^{\mathrm{gp}}$. Inverting a finite set of generators for this rational vector space gives a finitely presented sharp localization of P. We write P' for this sharp localization and show that $P' = \overline{P}$.

Clearly, the sharp localization $P \to P'$ factors $P \to \overline{P}$. Suppose that $x \in P$ projects to 0 in Q. Then x is an invertible element of $P + R$. That is, there are some $y \in P$ and $z \in R$ such that $x + y + z = 0$. Then

$x+y=-z\in -R\cap P$. By construction, $x+y$ projects to 0 in P' and, since $0\leq x\leq x+y$ in P, it follows that x also projects to 0 in P'. Thus every element of P that vanishes in $\overline{P}$ also vanishes in P', and we conclude $P'=\overline{P}$. □

Lemma 1.3.4. *Let M be a monoid containing an element y. Then the kernel of $M^{\mathrm{gp}}\to (M[-y]^\sharp)^{\mathrm{gp}}$ consists of all $x\in M^{\mathrm{gp}}$ such that $x\prec y$.*

Proof. Let N be the non-sharp, saturated submonoid of M^{gp} generated by M and $-y$, and suppsoe $x\in M^{\mathrm{gp}}$. Then x projects to 0 in $(M[-y]^\sharp)^{\mathrm{gp}}$ if and only if both x and $-x$ lie in N. This happens if only if there are positive integers m and n such that $x+ny\in M$ and $-x+my\in M$, or, in other words, if and only if $-ny\leq x\leq my$. □

Lemma 1.3.5. *Let $f: M\to N$ be a monoid homomorphism and suppose $x\in M^{\mathrm{gp}}$ and $z\in N$. Let $\overline{N}=N[-z]^\sharp$ and let $\overline{M}$ be the sharpening of $M\to\overline{N}$. Then the image of x in $\overline{M}^{\mathrm{gp}}$ vanishes if and only if there is a $y\in M$ such that $x\prec y$ and $f(y)\prec z$.*

Proof. It is clear that if there is a $y\in M$ such that $x\prec y$ and $f(y)\prec z$ then x projects to 0 in $\overline{M}$. Conversely, Lemma 1.3.4 shows that x projects to 0 in $\overline{M}$ if and only if there is some positive $y\in M$ such that $x\prec y$ and y projects to 0 in $\overline{M}$. But a positive y projects to 0 in $\overline{M}$ if and only if it projects to 0 in $\overline{N}$ if and only if $f(y)\prec z$, by a second application of Lemma 1.3.4. □

Lemma 1.3.6. *The base change of an infinitesimal morphism is infinitesimal.*

Proof. Suppose that $f: M\to N$ is an infinitesimal homomorphism of monoids. Let $N'\to N$ be a monoid homomorphism, and let $f': M'\to N'$ and $g: M'\to M$ be the morphisms induced by base change. Suppose that $x'\in M'^{\mathrm{gp}}$ and $y'\in M'$ and $f'(x')\in N'$ and $x'\prec y'$ and $f'(y')\prec f'(x')$. Let $x=g(x')$ and $y=g(y')$. Then $f(x)$ is the image of $f'(x')$, hence lies in N and $x\prec y$ and $f(y)\prec f(x)$. Therefore $x\in M$ because f is infinitesimal. We conclude that $x'\in M'$ by definition of a fiber product. □

Lemma 1.3.7. *Suppose that $P\to Q$ is birhizomic at $Q\to M$ and that $M\to N$ is a monoid homomorphism. Then $P\to Q$ is also birhizomic at $Q\to N$.*

Proof. Immediate from Lemma 1.3.6. □

Lemma 1.3.8. *If $f: P\to Q$ is birhizomic at $g: Q\to M$ and $P'\to Q'\to M'$ is induced by pushout from f and g along a homomorphism $P\to P'$, then $P'\to Q'$ is birhizomic at $Q'\to M'$.*

Proof. Immediate from Lemma 1.3.7, the formulation of birhizomicity as a lifting property, and the fact that finite presentation is preserved by pushout. □

Lemma 1.3.9. *A family of morphisms $P\to Q_i$ is a birhizomic cover if and only if, for every* sharp *valuation $P\to V$ there is a factorization through some Q_i such that $P\to Q_i$ is birhizomic at $Q_i\to V$.*

Proof. Clearly, if every valuation lifts to a birhizomic point then every sharp valuation does. Conversely, suppose that sharp valuations of P lift to birhizomic points. Let $P\to V$ be an arbitrary valuation. It can be factored through a sharp valuation $P\to W$ [Ogu18, Proposition I.2.4.1],[1] which factors through some Q_i where $P\to Q_i$ is birhizomic. But then $P\to Q_i$ is also birhizomic at $Q_i\to V$ by Lemma 1.3.7. □

Proposition 1.3.10. *Let $P\to Q$ be a sharp monoid homomorphism of finite presentation. Let K be the kernel of $P^{\mathrm{gp}}\to Q^{\mathrm{gp}}$. Then K is torsion-free and $\dim \mathbf{Q}\otimes K<\infty$.*

[1] The statement of [Ogu18, Proposition I.2.4.1] says that if L is an abelian group containing P then any submonoid of L that contains P and is maximal with respect to domination is valuative. In fact, the proof only requires that L be valuative as a not-necessarily-sharp monoid. We apply this version of [Ogu18, Proposition I.2.4.1] with L being the preïmage of V under $P^{\mathrm{gp}}\to V^{\mathrm{gp}}$.

Proof. The torsion-free assertion is immediate because $K \subseteq P^{\mathrm{gp}}$ and P is saturated.

One can make $P \to Q$ surjective by adding a finite number of generators to P. This may increase the rank of K, but only by a finite amount. We therefore assume without loss of generality that $P \to Q$ is surjective.

For each finitely generated subgroup $K_\alpha \subseteq K$, let R_α be the sharpening of the saturation of the image of $P \to P^{\mathrm{gp}}/K_\alpha$. Since the finitely generated subgroups of K are filtered with respect to inclusion, the R_α form a filtered system, and, moreover, $\operatorname{colim}_\alpha R_\alpha = Q$. Since $P \to Q$ is of finite presentation, the identity homomorphism $Q \to Q$ factors through some R_α by a homomorphism commuting with the maps from P. The factorization implies $R_\alpha^{\mathrm{gp}} = Q^{\mathrm{gp}}$ as quotients of P^{gp}, and therefore that K is the kernel of $P^{\mathrm{gp}} \to R_\alpha^{\mathrm{gp}}$. But $x \in P^{\mathrm{gp}}$ projects to 0 in R_α^{gp} if and only if it is torsion in P^{gp}/K_α, and therefore if and only if $x \in \mathbf{Q} \otimes K_\alpha$. That is, $\mathbf{Q} \otimes K = \mathbf{Q} \otimes K_\alpha$ and, in particular, $\mathbf{Q} \otimes K$ has finite dimension. □

1.4 Reinterpretation of birhizomicity

Lemma 1.4.1. *Let N be a monoid and let L be a torsion-free abelian group. Let $M \subset N \times L$ be the set of pairs (a, b) where $a \neq 0$ or $b = 0$. Then M is a monoid an the projection $M \to N$ is infinitesimal.*

Proof. We leave to the reader to verify that M is a(n integral saturated, sharp) monoid. Let $f : M \to N$ be the projection. Suppose $x \in M^{\mathrm{gp}}$ and $y \in M$ and $x \prec y$ and $f(x) \in N$ and $f(y) \prec f(x)$. If $f(x) \neq 0$ then $x \in M$ by definition and if $f(x) = 0$ then $f(y) = 0$ so $y = 0$ by definition, and therefore $x = 0$. □

Lemma 1.4.2. *Let $P \to Q \to N$ be sharp monoid homomorphisms. If $P \to Q$ is birhizomic at $Q \to N$ then $P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$ is bijective.*

Proof. Let $M \subset N \times P^{\mathrm{gp}} \to N$ be set of (a, b) such that $a \neq 0$ or $b = 0$. Then $M \to N$ is an infinitesimal monoid homomorphism by Lemma 1.4.1. Let $P \to M \subset N \times P^{\mathrm{gp}}$ be given by $P \to N$ on the first factor and the inclusion on the second factor. Sharpness of $P \to N$ guarantees factorization through M. Then lifting in the diagram

$$\begin{array}{ccc} P & \longrightarrow & M \\ \downarrow & \nearrow & \downarrow \\ Q & \longrightarrow & N \end{array} \tag{1.1}$$

shows that $P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$ is a split injection.

Let $L = Q^{\mathrm{gp}}/P^{\mathrm{gp}}$ and let $M \subset N \times L$ be the set of pairs (a, b) such that $a \neq 0$ or $b = 0$. Note that L is torsion-free because $P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$ is a split injection. Then M is a monoid and $M \to N$ is infinitesimal by Lemma 1.4.1. Uniqueness of lifting in a diagram (1.1) shows that $Q \to L$ vanishes, hence that $P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$ is surjective. □

Lemma 1.4.3. *Every birhizomic cover $\{P \to Q_i\}$ has a subcover where all $P^{\mathrm{gp}} \to Q_i^{\mathrm{gp}}$ are bijective (and in particular are sharp).*

Proof. Since every valuation is refined by a sharp valuation, we can refine our cover to include only those that have birhizomic lifts of sharp valuations by Lemma 1.3.9. In this case, every $P \to Q_i$ is sharp.

For each valuation $v : P \to V$ of P, choose a lift to some Q_i where $P \to Q_i$ is birhizomic. Let Q_v be the sharpening of $Q_i \to V$. Then one verifies easily using Lemma 1.3.1 that the maps $P \to Q_v$ are a birhizomic cover refining $\{P \to Q_i\}$. But now $P \to Q_v \to V$ are sharp homomorphisms, so $P^{\mathrm{gp}} \to Q_v^{\mathrm{gp}}$ is bijective by Lemma 1.4.2. □

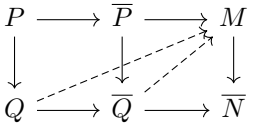
Let $P \to N$ be a monoid homomorphism. We write $P_\sharp$ for the sheaf on $\operatorname{Spec} N$ whose value on a sharp localization $\overline{N}$ of N is the sharpening of $P \to \overline{N}$, and we write $P_\sharp^{\mathrm{gp}}$ for the associated sheaf of groups. If

$$P \to Q \to N$$

are monoid homomorphisms, we write $\mathcal{L}_{Q/P}\big|_N$ for the complex $[P_\sharp^{\mathrm{gp}} \to Q_\sharp^{\mathrm{gp}}]$, in degrees $[-1, 0]$, on $\operatorname{Spec} N$.

Theorem 1.4.4. *Let $P \to Q \to N$ be monoid homomorphisms. Assume that $P \to Q$ is of finite presentation. Then $P \to Q$ is birhizomic at $Q \to N$ if and only if $\mathcal{L}_{Q/P}\big|_N$ is acyclic.*

Proof. Suppose first that $P \to Q$ is birhizomic at $Q \to N$. Suppose $N \to \overline{N}$ is a sharp localization of N and that $\overline{P} \to \overline{Q}$ is induced by sharpening $P \to \overline{N}$ and $Q \to \overline{N}$. Then Lemma 1.3.2 shows that $\overline{P} \to \overline{Q}$ is of finite presentation. If



is a lifting problem, with $M \to \overline{N}$ infinitesimal, the longer dashed arrow exists because $P \to Q$ is birhizomic at $Q \to \overline{N}$, by Lemma 1.3.7. Then a unique factorization through $\overline{Q}$ exists because if $x \in Q$ projects to 0 in $\overline{N}$, it also projects to 0 in M since $M \to \overline{N}$ is sharp, by Lemma 1.3.1. The square on the right commutes because $P \to \overline{P}$ and $Q \to \overline{Q}$ are epimorphisms. We conclude that $\overline{P} \to \overline{Q}$ is birhizomic, and therefore that $\overline{P}^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$ is a bijection. This shows that $\mathcal{L}_{Q/P}\big|_N$ is acyclic.

Conversely, suppose that $\mathcal{L}_{Q/P}\big|_N$ is acyclic. Let $g : M \to N$ be an infinitesimal homomorphism and let $P \to M$ lift $P \to N$. We can replace P and Q with the sharpenings of $P \to N$ and $Q \to N$, respectively, since $M \to N$ is sharp by Lemma 1.3.1. We know then that $P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$ is bijective, so composing with an inverse gives $f : Q^{\mathrm{gp}} \to M^{\mathrm{gp}}$. This already shows a lift is unique if it exists. For existence, we must show that $f(Q) \subset M$. Suppose $x \in Q$. Then $g(f(x)) \in N$. Let $\overline{N} = N[-g(f(x))]^\sharp$ and let $\overline{M}$ (resp. $\overline{P}$, resp. $\overline{Q}$) be the sharpening of $M \to \overline{N}$ (resp. $P \to \overline{N}$, resp. $Q \to \overline{N}$). Then x projects to 0 in $\overline{Q}^{\mathrm{gp}}$, hence also in $\overline{P}^{\mathrm{gp}}$, since $\overline{P}^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$ is an isomorphism. It follows that $f(x)$ projects to 0 in $\overline{M}^{\mathrm{gp}}$, hence $f(x)$ lies in M, by Lemma 1.3.5, since $M \to N$ is infinitesimal. □

Proposition 1.4.5. *Let $P \xrightarrow{f} Q \to M \xrightarrow{g} N$ be monoid homomorphisms, with g exact. Then $P \to Q$ is birhizomic at $Q \to M$ if and only if it is birhizomic at $Q \to N$.*

Proof. If $P \to Q$ is birhizomic at $Q \to M$ then it is birhizomic at $Q \to N$ by Lemma 1.3.7. By Theorem 1.4.4, the map $P \to Q$ is birhizomic at $Q \to M$ if and only if, when $\overline{M}$ is a sharp localization of M, and $\overline{P}$ and $\overline{Q}$ denote the sharpenings of $P \to \overline{M}$ and $Q \to \overline{M}$, then $\overline{P}^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$ is an isomorphism. The same criterion holds for birhizomicity at $Q \to N$, where $\overline{N}$ is the pushout of $M \to \overline{M}$ along $M \to N$. But $\overline{M} \to \overline{N}$ is the pushout of an exact morphism, hence is exact [Tsu19, Proposition I.3.2 (2)]. Therefore the sharpenings of $P \to \overline{N}$ and $P \to \overline{M}$ coincide. Likewise, the sharpenings of $Q \to \overline{N}$ and $Q \to \overline{M}$ coincide. Thus f is birhizomic at $Q \to N$ if and only if it is birhizomic at $Q \to M$. □

1.5 Finite type approximation

Lemma 1.5.1. *Suppose that P is a monoid and Q is generated relative to P by elements $f_1, \ldots, f_n$. Let $Q \to \overline{Q}$ be a sharp localization and let $P \to \overline{P}$ be the sharpening of $P \to \overline{Q}$. If $\sum a_i f_i$ restricts to 0 in $\overline{Q}^{\mathrm{gp}}/\overline{P}^{\mathrm{gp}}$, for some integers a_i, then either $\sum a_i f_i = 0$ in $Q^{\mathrm{gp}}/P^{\mathrm{gp}}$ or there is some f_i that is inverted in $\overline{Q}$.*

Proof. Assume that $\sum a_i f_i$ restricts to 0 in $\overline{Q}^{\mathrm{gp}}/\overline{P}^{\mathrm{gp}}$ and that no f_i is inverted in $\overline{Q}$. Since $\overline{Q}$ is a filtered colimit of finitely presented sharp localizations, we may assume that $\overline{Q}$ is itself a finitely presented sharp localization. That is, $\overline{Q} = Q[-h]^\sharp$ for some $h \in Q$. We can write $h = g + \sum b_i f_i$ with $g \in P$ and all $b_i \geq 0$. But if any b_i is nonzero then $0 \leq f_i \leq h$ in Q and therefore f_i would localize to 0 in $\overline{Q}$. Therefore $b_i = 0$ for all i and $h \in P$.

Since $\sum a_i f_i$ localizes to an element of $\overline{P}^{\mathrm{gp}}$, and since $P^{\mathrm{gp}} \to \overline{P}^{\mathrm{gp}}$ is surjective, there is some $g \in P^{\mathrm{gp}}$ such that $g + \sum a_i f_i$ projects to 0 in $\overline{Q}^{\mathrm{gp}}$. Therefore $g + \sum a_i f_i \prec h$. Replacing h by nh for some integer $n \geq 1$, this means that

$$-h \leq g + \sum a_i f_i \leq h$$

and therefore that there are elements $u, v \in Q$ with

$$-h + u = g + \sum a_i f_i = h - v.$$

But h projects to 0 in $\overline{Q}^{\mathrm{gp}}$, so $u + v$ projects to 0 in $\overline{Q}^{\mathrm{gp}}$. We can write $u = u_0 + \sum c_i f_i$ and $v = v_0 + \sum d_i f_i$ where u_0 and v_0 are in P and all c_i and d_i are non-negative integers. By the same reasoning as above, we must have all $c_i = 0$ and $d_i = 0$ since none of the f_i is inverted in $\overline{Q}$. Therefore u and v are both in P. But this means that $\sum a_i f_i \in P^{\mathrm{gp}}$, and therefore $\sum a_i f_i$ represents 0 in $Q^{\mathrm{gp}}/P^{\mathrm{gp}}$. □

Lemma 1.5.2. *Suppose that $u : P \to Q$ is an injective homomorphism of monoids and Q is generated relative to P by elements $f_1, \ldots, f_n$. Let $Q \to \overline{Q}$ be a sharp localization and let $P \to \overline{P}$ be the sharpening of $P \to \overline{Q}$. If $\overline{P}^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$ is not injective then there is some f_i that is inverted in $\overline{Q}$.*

Proof. Since any sharp localization $\overline{Q}$ of Q is a filtered colimit of finitely presented sharp localizations, it is sufficient to assume that $\overline{Q}$ is finitely presented relative to Q. We may therefore present $\overline{Q}$ as $Q[-g]^\sharp$. We can write $g = h + \sum a_i f_i$, where the a_i are non-negative integers and $h \in P$. If any of the a_i is nonzero then f_i will be inverted in $Q[-g]^\sharp$. Therefore, we assume that $a_i = 0$ for all i. That is, $g \in P$. If $x \in P^{\mathrm{gp}}$ represents an element of the kernel of $\overline{P}^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$, then $x \prec g$. That is, there are elements $y + \sum b_i f_i$ and $z + \sum c_i f_i$ such that

$$-g + y + \sum b_i f_i = x = g - z - \sum c_i f_i$$

in Q^{gp}. But g projects to 0 in $\overline{Q}$, so $y + z + \sum (b_i + c_i) f_i$ also projects to 0. By assumption, none of the f_i projects to 0, so we must have $b_i = c_i = 0$. Therefore $-g \leq x \leq g$ in P (and not merely in Q). Thus $x \prec g$ in P and therefore x projects to 0 in $\overline{P}^{\mathrm{gp}}$. □

Corollary 1.5.3. *Suppose that P is a monoid and Q is generated relative to P by elements $f_1, \ldots, f_n$. Let $Q \to \overline{Q}$ be a sharp localization and let $P \to \overline{P}$ be the sharpening of $P \to \overline{Q}$. If*

$$H^0 \mathcal{L}_{Q/P}(Q) \to H^0 \mathcal{L}_{Q/P}(\overline{Q})$$

is not an isomorphism or

$$\mathbf{Q} \otimes H^{-1} \mathcal{L}_{Q/P}(Q) \to \mathbf{Q} \otimes H^{-1} \mathcal{L}_{Q/P}(\overline{Q})$$

is not surjective, then some f_i is inverted in $\overline{Q}$.

Proof. Lemma 1.5.1 shows directly that if

$$H^0 \mathcal{L}_{Q/P}(Q) \to H^0 \mathcal{L}_{Q/P}(\overline{Q})$$

is not an isomorphism then some f_i is inverted in $\overline{Q}$.

For the claim about H^{-1}, we replace P with the sharpening of $P \to Q$, which does not change $\mathcal{L}_{Q/P}$, and assume without loss of generality that $P \to Q$ is sharp. Let P_1 be the saturated image of P in Q. Let K be the kernel of $P^{\mathrm{gp}} \to P_1^{\mathrm{gp}}$. Let J be the saturated image of K in $\overline{P}^{\mathrm{gp}}$ and let $\overline{P}_1$ be the saturated image of $\overline{P}$ in $\overline{P}^{\mathrm{gp}}/J$ (in other words, the pushout of $P \to P_1$ along $P \to \overline{P}$). Then $\overline{P}_1 \to \overline{Q}$ is sharp : if $x \in \overline{P}_1$ maps to 0 in $\overline{Q}$ then nx is the image of $y \in \overline{P}$ for some positive integer n, and y projects to 0 in $\overline{Q}$, hence vanishes, since $\overline{P} \to \overline{Q}$ is sharp.

Let $\overline{K}$ be the kernel of $\overline{P}^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$. Then $\overline{K}/J$ is the kernel of $\overline{P}_1^{\mathrm{gp}} \to \overline{Q}^{\mathrm{gp}}$. If $\mathbf{Q} \otimes K \to \mathbf{Q} \otimes \overline{K}$ is not surjective then $\overline{K}/J \neq 0$ and therefore Lemma 1.5.2 implies that some f_i is inverted in $\overline{Q}$. □

Corollary 1.5.4. *Suppose that $P \to Q$ is a homomorphism of monoids, that $P' \to Q'$ is induced by pushout along a homomorphism $P \to P'$, and that $\overline{P}'$ is the sharpening of $P' \to Q'$. Fix a finite set of generators for Q relative to P. Let K be the kernel of $P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$ and let $\overline{K}$ be the kernel of $\overline{P}'^{\mathrm{gp}} \to Q'^{\mathrm{gp}}$. Then $\mathbf{Q} \otimes K \to \mathbf{Q} \otimes \overline{K}$ is surjective or there is some generator of Q that is inverted in Q'.*

Proof. Since P' is a filtered colimit of monoids that are finitely presented relative to P, it will be sufficient to assume that P' is finitely presented relative to P. We can present P' by repeatedly adding a free generator, or repeatedly adding elements of P^{gp} and sharpening. (To see this, note that any relation $x = 0$, for $x \in P^{\mathrm{gp}}$, can be imposed by adjoining x and $-x$ to P and sharpening.) Since compositions of surjections are surjective, we can consider these two operations in isolation. If $P' = P \oplus \mathbf{N}$ then $P'^{\mathrm{gp}} = P^{\mathrm{gp}} \oplus \mathbf{Z}$ and $Q'^{\mathrm{gp}} = Q^{\mathrm{gp}} \oplus \mathbf{Z}$; the map $P' \to Q'$ is sharp, and $K \to \overline{K}$ is an isomorphism. If $P' = P[x]$ for some $x \in P^{\mathrm{gp}}$ then $Q' = Q[x]^{\mathrm{sat}\sharp}$. If $-x \notin Q$, then $(Q[x]^{\mathrm{sat}\sharp})^{\mathrm{gp}} = Q^{\mathrm{gp}}$ and $(P[x]^{\mathrm{sat}\sharp})^{\mathrm{gp}} = P^{\mathrm{gp}}$ so $K \to \overline{K}$ is an isomorphism. If $-x \in Q$ then $Q[x]^{\mathrm{sat}\sharp}$ is a sharp localization of Q and Corollary 1.5.3 shows that, with $\overline{P}$ denoting the sharpening of $P \to Q'$, that either some generator of Q is inverted in Q' or K surjects onto the kernel of $\overline{P}^{\mathrm{gp}} \to Q'^{\mathrm{gp}}$. Since $\overline{P}'^{\mathrm{gp}}$ is a quotient of $\overline{P}^{\mathrm{gp}}$ in this case, K also surjects onto the kernel of $\overline{P}'^{\mathrm{gp}} \to Q'^{\mathrm{gp}}$. □

Theorem 1.5.5. *Let $P_\alpha \to Q_\alpha \to M_\alpha$ be a family of monoid homomorphisms indexed by a filtered diagram with $P_\alpha \to Q_\alpha$ of finite presentation. For each $\alpha \to \beta$, suppose the commutative diagram*

$$\begin{array}{ccccc} P_\alpha & \longrightarrow & Q_\alpha & \longrightarrow & M_\alpha \\ \downarrow & & \downarrow & & \downarrow \\ P_\beta & \longrightarrow & Q_\beta & \longrightarrow & M_\beta. \end{array}$$

exhibits $Q_\beta = Q_\alpha \oplus_{P_\alpha} P_\beta$ as the pushout among (sharp, integral, saturated) monoids. Write $P \to Q \to M$ for the colimit of the system.

If $P \to Q$ is birhizomic at $Q \to M$, then there is an index α such that $P_\alpha \to Q_\alpha$ is birhizomic at $Q_\alpha \to M_\alpha$.

Proof. Abbreviate $H^0_\alpha = H^0\mathcal{L}_{Q_\alpha/P_\alpha}\big|_{M_\alpha}$ and $H^{-1}_\alpha = H^{-1}\mathcal{L}_{Q_\alpha/P_\alpha}\big|_{M_\alpha}$.

Choose a finite set $\{f_1, \cdots, f_n\}$ of generators for some Q_{α_0} over P_{α_0}. Replacing the indexing diagram, we may assume without loss of generality that α_0 is initial in the diagram, so that $f_1, \ldots, f_n$ generate every Q_α relative to P_α. For each subset $I \subset \{f_1, \ldots, f_n\}$, let $U(I)_\alpha$ be the open subset of $\operatorname{Spec} M_\alpha$ where the images of all $f \in I$ in M_α vanish.

We argue that there is some index α such that, for every sharp localization $\overline{M}$ of M_α, the groups $H^0_\alpha(\overline{M})$ and $\mathbf{Q} \otimes H^{-1}_\alpha(\overline{M})$ vanish. The latter implies the vanishing of $H^{-1}_\alpha(\overline{M})$ because $H^{-1}_\alpha(\overline{M})$ is torsion-free (since it is a subgroup of the associated group of the sharpening of $P \to \overline{M}$, which is torsion-free). This implies that $P_\alpha \to Q_\alpha$ is birhizomic at $Q_\alpha \to M_\alpha$ by Theorem 1.4.4.

Let I be the set of indices i such that f_i is inverted in $\overline{M}$. Then by Corollary 1.5.3, the maps $H^0_\alpha(U(I)_\alpha) \to H^0_\alpha(\overline{M})$ and $\mathbf{Q} \otimes H^{-1}_\alpha(U(I)_\alpha) \to \mathbf{Q} \otimes H^{-1}_\alpha(\overline{M})$ are surjective. Therefore it suffices to find an α such that $H^0_\alpha(U(I)_\alpha)$ and $\mathbf{Q} \otimes H^{-1}_\alpha(U(I)_\alpha)$ both vanish for all I.

Since $H^0_\alpha(U(I)_\alpha)$ is finitely generated as an abelian group, and $\mathbf{Q} \otimes H^{-1}_\alpha(U(I)_\alpha)$ is a finite dimensional $\mathbf{Q}$-vector space (by Proposition 1.3.10 and Lemma 1.3.2), and since $\operatorname{colim} H^i_\alpha(U(I)_\alpha) = 0$ because $P \to Q$ is birhizomic at $Q \to M$, it will be sufficient to show that $H^0_\alpha(U(I)_\alpha) \to H^0_\beta(U(I)_\beta)$ and $\mathbf{Q} \otimes H^{-1}_\alpha(U(I)_\alpha) \to \mathbf{Q} \otimes H^{-1}_\beta(U(I)_\beta)$ are surjective for all sufficiently large values of α and β. Because filtered colimits commute with kernels, cokernels, sharpening, and passage to the associated group, $H^i\mathcal{L}_{Q/P}(U(I))$ can be identified with $\operatorname{colim} H^i_\alpha(U(I)_\alpha)$, and the vanishing of the filtered colimit implies the vanishing at some finite stage.

For H^0 this is clear. For H^{-1}, Corollary 1.5.4 shows that $\mathbf{Q} \otimes H^{-1}_\alpha(U(I)_\alpha) \to \mathbf{Q} \otimes H^{-1}_\beta(U(I)_\beta)$ is surjective unless some generator of $\overline{Q}_\alpha$ relative to $\overline{P}_\alpha$ is inverted in $\overline{Q}_\beta$ (with $\overline{P}_\alpha = (P_\alpha)_\sharp(U(I)_\alpha)$, with $\overline{Q}_\alpha = (Q_\alpha)_\sharp(U(I)_\alpha)$, and with $\overline{Q}_\beta = (Q_\beta)_\sharp(U(I)_\beta)$). Since $\{f_1, \ldots, f_n\}$ is a finite set, every element of $\{f_1, \ldots, f_n\}$ that is inverted in $Q_\sharp(U(I))$ has already been inverted in some $(Q_\alpha)_\sharp(U(I)_\alpha)$. □

Corollary 1.5.6. *Suppose that $P \to Q \to N$ is a sequence of monoid homomorphisms, and that $P \to Q$ is birhizomic at $Q \to N$. Then there is a finitely generated monoid P_0 and monoid homomorphism $P_0 \to Q_0$ such that $P \to Q$ is induced from $P_0 \to Q_0$ by pushout along a homomorphism $P_0 \to P$, and $P_0 \to Q_0$ is birhizomic at $Q_0 \to N$.*

Proof. Since $P \to Q$ is of finite presentation, we can find a finitely generated monoid P', a homomorphism $P' \to P$, and a homomorphism $P' \to Q'$ of finite presentation that induces $P \to Q$ by pushout. We take $\{P_\alpha\}$ to be the system of all factorizations of $P' \to P$ through a monoid of finite presentation, and we define Q_α to be the pushout of Q' along $P' \to P_\alpha$. Then we apply Theorem 1.5.5 to the following family of diagrams :

$$\begin{array}{ccccc} P_\alpha & \longrightarrow & Q_\alpha & \longrightarrow & N \\ \downarrow & & \downarrow & & \| \\ P & \longrightarrow & Q & \longrightarrow & N \end{array}$$

□

1.6 Quasicompactness

Definition 1.6.1. Let A be a partially ordered abelian group. We write $|\mathrm{Cone}\, A|$ for the set of valuations of A. We say that a subset of $|\mathrm{Cone}\, A|$ is *constructible* if it can be described by a quantifier-free formula involving elements of A and $\leq$ and Boolean operations. The *constructible topology* is the coarsest topology on $|\mathrm{Cone}\, A|$ in which the constructible sets are open.

An order preserving homomorphism $A \to B$ induces a map $|\mathrm{Cone}\, B| \to |\mathrm{Cone}\, A|$ that carries constructible subsets to constructible subsets. If P is a monoid, we also write $|\mathrm{Cone}\, P| = |\mathrm{Cone}\, P^{\mathrm{gp}}|$, with P^{gp} given the partial order induced by P.

Proposition 1.6.2. *Let A be a partially ordered abelian group. Then $|\mathrm{Cone}\, A|$ is quasicompact in the constructible topology.*

Proof. Let $W = |\mathrm{Cone}\, A|$. We must show that every ultrafilter on W converges [Bou07, Définition I.9.1.1]. Let U be an ultrafilter on W. For each $x, y \in A$, we write $U(x) \leq U(y)$ if U contains $\{v \in W \mid v(x) \leq v(y)\}$. Let R be the set of pairs $(x, y) \in A \times A$ such that $U(x) \leq U(y)$. It is a straightforward verification, using the fact that U is an ultrafilter, that Rextends the order relation of A, hence gives a valuation w of A.

We can define $U(x) < U(y)$ to mean that U contains $\{v \in W \mid v(x) < v(y)\}$. We note that because U is an ultrafilter and the order $v \in W$ are total, the assertion $U(y) < U(x)$ is the logical negation of $U(x) \leq U(y)$.

To conclude, we show that U converges to w in the constructible topology. We must show that if K is any open neighborhood of w in the constructible topology on $|\mathrm{Cone}\, A|$ then $K \in U$. Any such K is a union of basic constructible sets, each of which is defined by the conjunction of finitely many inequalities. Since U is a filter, and in particular stable under finite intersection, it will suffice to assume that K is defined by a single inequality, say $x \leq y$ (resp. $x < y$). Since K is a neighborhood of w, we have $w(x) \leq w(y)$ (resp. $w(x) < w(y)$), which implies that $U(x) \leq U(y)$ (resp. $U(x) < U(y)$) by the definition of w, and therefore that $K \in U$, as required. □

Proposition 1.6.3. *Let $f : P \to Q$ be a homomorphism of monoids such that Q is of finite presentation relative to P. Then the image of $|\mathrm{Cone}(Q)| \to |\mathrm{Cone}(P)|$ is constructible.*

Proof. Let P' be the preïmage of Q under $f : P^{\mathrm{gp}} \to Q^{\mathrm{gp}}$. Then we claim the image of $\varphi : |\mathrm{Cone}(Q)| \to |\mathrm{Cone}(P)|$ is represented by $|\mathrm{Cone}(P')|$. Indeed, if $v : P \to V$ is a valuation then v lies in the image of φ if and only if v extends to a valuation that lifts to Q. Since the formation of P' commutes with pushout (see the proof of [Tsu19, Proposition I.3.2 (2)]), it is sufficient to verify the claim when $P = V$ and Q is replaced by the pushout Q_v along v. Since V is valuative, $|\mathrm{Cone}(Q_v)| \to |\mathrm{Cone}(V)|$ is surjective if and only if $V \to Q_v$ is sharp. Indeed, every monoid has a sharp valuation [Ogu18, Proposition I.2.4.1], so the identity valuation of V lifts, and therefore every valuation lifts by composition. But a homomorphism whose domain is valuative is sharp if and only if it is exact so $|\mathrm{Cone}(Q_v)| \to |\mathrm{Cone}(V)|$ is surjective if and only if $P'_v = V$.

We therefore only need to show that P' is finitely presented relative to P. But f is of finite presentation, so there is a monoid P_0 of finite type and a homomorphism $P_0 \to Q_0$ such that f is induced by pushout along a homomorphism $P_0 \to P$. Again because the formation of P' is stable under pushout, it is sufficient to show that P'_0 is finitely presented relative to P_0.

To conclude we note that because P_0 and Q_0 are finitely generated, they can be viewed as rational polyhedral cones inside the real vector spaces $\mathbf{R} \otimes P_0^{\mathrm{gp}}$ and $\mathbf{R} \otimes Q_0^{\mathrm{gp}}$. In particular, Q_0 is defined by finitely many integral inequalities, so its preïmage P'_0 is likewise defined by finitely many integral inequalities. Thus P'_0 is the set of integral points of a rational polyhedral cone, so it is finitely presented. □

Proposition 1.6.4. *Let $P \to Q \to V$ be a sequence of monoid homomorphisms with V valuative and $P \to Q$ birhizomic at $Q \to V$. Then there is a constructible subset U of $|\mathrm{Cone}(Q)|$ containing $P \to V$ such that $P \to Q$ is birhizomic at every $Q \to W$ in U.*

Proof. We can present V as a filtered colimit of monoids M_α with each M_α of finite presentation over Q. We take $P_\alpha = P$ and $Q_\alpha = Q$ for all α and apply Theorem 1.5.5. This gives an index α such that $P \to Q$ is birhizomic at $Q \to M_\alpha$. Let M'_α be the preïmage of M_α under $Q^{\mathrm{gp}} \to M_\alpha^{\mathrm{gp}}$. Then $M'_\alpha \to M_\alpha$ is exact, so $P \to Q$ is also birhizomic at $Q \to M'_\alpha$ by Proposition 1.4.5. Then $P \to Q$ is birhizomic at every $Q \to W$ in $|\mathrm{Cone}\, M'_\alpha| \subset |\mathrm{Cone}\, Q|$ by Lemma 1.3.7. Finally, $|\mathrm{Cone}\, M'_\alpha| \subset |\mathrm{Cone}\, Q|$ is constructible by Proposition 1.6.3. □

Corollary 1.6.5. *Every birhizomic cover of a monoid P has a finite subcover.*

Proof. Suppose that $\mathcal{U} = \{P \to Q_i\}$ is a birhizomic cover. By Proposition 1.6.4, the set of valuations of Q_i at which $P \to Q_i$ is birhizomic is a union of images of $|\mathrm{Cone}(W)|$ for monoids W that are finitely presented relative to Q_i, hence finitely presented relative to P. Each $|\mathrm{Cone}(W)|$ projects to a constructible subset of $|\mathrm{Cone}(P)|$ by Proposition 1.6.3. Since $|\mathrm{Cone}\, P|$ is quasicompact, there is a finite subset of the $|\mathrm{Cone}(W)|$ that covers $|\mathrm{Cone}(P)|$, and therefore there is a finite subset of the $P \to Q_i$ that already forms a birhizomic cover of P. □

Proposition 1.6.6. *Suppose that $P_\alpha \to W_{\alpha,i}$ is a cocartesian filtered system of monoid homomorphisms of finite presentation with colimits $P \to W_i$. Assume that the $|\mathrm{Cone}(W_i)|$ cover $|\mathrm{Cone}(P)|$. Then there is some index α such that the $|\mathrm{Cone}(W_{\alpha,i})|$ cover $|\mathrm{Cone}(P_\alpha)|$.*

Proof. We may assume that the α are indexed by a filtered partially ordered set (rather than a filtered category). Suppose for the sake of contradiction that, for every α, there is a valuation $v_\alpha : P_\alpha \to V_\alpha$ that does not extend to any of the $W_{\alpha,i}$. Let U be an ultrafilter on the set of indices $\{\alpha\}$ extending the order filtration. Let

$$\prod_U V_\alpha^{\mathrm{gp}} = \operatorname*{colim}_{S \in U} \prod_{\alpha \in S} V_\alpha^{\mathrm{gp}}$$

be the ultraproduct of the V_α^{gp}. This is a totally ordered abelian group, so the non-negative elements V are a valuative monoid. We obtain a valuation $v : P \to V$ by sending $x \in P_\alpha$ to $(v_\beta(i_{\alpha\beta}(x)))_\beta$ where $i_{\alpha\beta} : P_\alpha \to P_\beta$ is the map associated with every $\alpha \leq \beta$. Moreover, v cannot lie in the image of any $|\mathrm{Cone}(W_i)|$. Indeed, by Proposition 1.6.3, there is a quantifier-free formula F such that v lies in the image of $|\mathrm{Cone}(W_i)|$ if and only if $v(F)$ holds. This formula involves only finitely many elements of P and relations among them, so it makes sense and characterizes the image of $|\mathrm{Cone}(W_{\alpha,i})|$ in all sufficiently large P_α. If this formula holds in $\mathrm{colim}_{S \in U} \prod_{\alpha \in S} V_\alpha$, then Łoś's theorem implies that there is some $S \in U$ such that it also holds in V_α for all $\alpha \in S$. Taking the projection on any $\alpha \in S$, we find that $v_\alpha(F)$ also holds, and therefore that v_α lifts to $W_{\alpha,i}$, contrary to our original assumption. □

1.7 The intersection complex

Definition 1.7.1. Let $\mathcal{U} = \{P \to Q_i\}$ be a finite family of monoid homomorphisms indexed by $i \in I$ and let $T \subseteq \mathrm{Spec}\, P$ be a subset. We say that $\mathcal{U}$ *intersects* in T if the preimage of $T \subseteq \mathrm{Spec}\, P$ under the map

$$\mathrm{Spec}\Big(\bigoplus_P Q_i\Big)^\sharp \longrightarrow \mathrm{Spec}\, P$$

is nonempty.

Recall the center of a valuation $P \to V$ is the image of the closed point of $\mathrm{Spec}\, V$ in $\mathrm{Spec}\, P$.

Proposition 1.7.2. *A finite family $\{P \to Q_i\}$ intersects in T if and only if there is a valuation of P with center in T that lifts to every Q_i.*

Proof. Let t be a point of T. Assume that there is a valuation v of P centered at t that lifts to a valuation w_i of each Q_i. Let W be a compositum [Pie72, Corollary 2.2] of the value monoids of the w_i so that all w_i can be viewed as valued in the same valuative monoid. Then we obtain a valuation w of $\big(\bigoplus_P Q_i\big)^\sharp$ by the universal property of the pushout, and this projects to v by construction. The center of w is a point of $\mathrm{Spec}\big(\bigoplus_P Q_i\big)^\sharp$ lying in the fiber over t.

Now assume that the fiber of $\mathrm{Spec}\big(\bigoplus_P Q_i\big)^\sharp$ over t is nonempty. Let $P \to P'$ be the sharp localization that inverts all elements of P outside the prime ideal corresponding to t. For each i, let Q_i' be the induced sharp localization of Q_i. Then the fiber $\mathrm{Spec}\big(\bigoplus_P Q_i\big)^\sharp$ over t is the same as the fiber of $\mathrm{Spec}\big(\bigoplus_{P'} Q_i'\big)^\sharp$ over the closed point of $\mathrm{Spec}\, P$, hence the latter is nonempty. Choose a sharp valuation of $\big(\bigoplus_{P'} Q_i'\big)^\sharp$ using [Ogu18, Proposition I.2.4.1]. This projects to valuations of the Q_i' that project to the same sharp valuation of P'. Composing, these give valuations of the Q_i that project to a valuation of P centered at t. □

Construction 1.7.3. Let $\mathcal{U} = \{P \to Q_i\}$ be a family of monoid homomorphisms and $T \subseteq \operatorname{Spec} P$ a subset. We define

$$X(\mathcal{U}, T) \coloneqq \{\mathcal{V} \subseteq \mathcal{U} \;\big|\; \mathcal{V} \text{ intersects in } T\}.$$

We observe that if $\mathcal{V} \subset \mathcal{U}$ intersects in T, so does every subset of $\mathcal{V}$. Therefore $X(\mathcal{U}, T)$ is a simplicial complex. Our convention is that $X(\mathcal{U}, T) = \varnothing$ when $T = \varnothing$.

Proposition 1.7.4. *Let $\mathcal{U} = \{P \to Q_i\}$ be a family of monoid homomorphisms of finite presentation such that each map $P^{\mathrm{gp}} \to Q_i^{\mathrm{gp}}$ is surjective. Let $T \subseteq \operatorname{Spec} P$ be a nonempty, constructible closed subset (in the Zariski topology of $\operatorname{Spec} P$). Suppose each valuation of P centered in T lifts to a valuation of some Q_i. Then $X(\mathcal{U}, T)$ is contractible.*

Proof. The simplicial complex $X(\mathcal{U}, T)$ is a filtered union over the simplicial complexes associated to finite subfamilies of $\{P \to Q_i\}$. Filtered colimits of contractible simplicial complexes are contractible (this follows from [Lur26, Tag 00ZE], for example, and the fact that simplicial spheres are finite complexes), so it suffices to show that $X(\mathcal{V}, T)$ is contractible for all sufficiently large finite subfamilies $\mathcal{V}$. The open subsets of $\operatorname{Spec} P$ can be described with quantifier- and negation-free formulas involving propositions $x \leq 0$ for $x \in P$. In particular, they correspond to constructible subsets of $|\mathrm{Cone}(P)|$, and therefore so does the closed set T. Proposition 1.6.2 tells us that this subset is quasicompact. Therefore, if $\mathcal{V}$ is any sufficiently large finite subfamily of $\mathcal{U}$, then every valuation of P centered in T lifts to some Q_i in $\mathcal{V}$. It will suffice to prove the proposition for such $\mathcal{V}$, so we assume from now on that $\mathcal{U}$ is finite.

We choose a minimal finite set $\{\beta_1, \ldots, \beta_N\} \subset P^{\mathrm{gp}}$ such that each Q_i is generated by some subset of the β_i. We induct on the total number N of required generators.

The base case is where every Q_i is a sharp localization of P. The sharp localizations $P \to Q_i$ give a Zariski cover of T. Since T is nonempty and closed, it contains the unique closed point of $\operatorname{Spec} P$. Therefore the maximal ideal of P is the preïmage of the maximal ideal of some Q_j, so $P = Q_j$ for some j. Thus $X(\{P \to Q_i\}, T)$ has a deformation retraction onto the vertex Q_j in this case.

In the inductive case, there is at least one Q_i such that $P \to Q_i$ is not a localization. After reordering, we assume this is Q_1. Reordering the β_i, we assume that Q_1 is presented as $P[\beta_1, \ldots, \beta_n]^{\mathrm{sat}\sharp}$ with $\beta_1, \ldots, \beta_n \in P^{\mathrm{gp}}$. Since Q_1 is not a localization of P, there is some β_j, with $1 \leq j \leq n$, such that $-\beta_j \notin P$. Relabel indices so that $j = 1$.

We define $P_+ = P[\beta_1]^{\mathrm{sat}\sharp}$ and $P_- = P[-\beta_1]^{\mathrm{sat}\sharp}$ and $P_0 = P[\pm\beta_1]^{\mathrm{sat}\sharp}$. For each $\epsilon \in \{+, -, 0\}$ and each index i, the pushout $(P_\epsilon \oplus_P Q_i)^\sharp$ can be generated by a subset of $\{\beta_2, \cdots, \beta_N\}$. The restriction of $\mathcal{U}$ to P_ϵ therefore requires strictly fewer generators than did $\mathcal{U}$. Likewise every valuation of P_ϵ with center in $T_\epsilon = T \times_{\operatorname{Spec} P} \operatorname{Spec} P_\epsilon$ (which is constructible and closed in $\operatorname{Spec} P_\epsilon$) lifts to some base change $(P_\epsilon \oplus_P Q_i)^\sharp$. We set $X_\epsilon = X(\mathcal{U}_\epsilon, T_\epsilon)$ where $\mathcal{U}_\epsilon = \{P_\epsilon \to (P_\epsilon \oplus_P Q_i)^{\mathrm{sat}\sharp}\}$.

We will argue below that $X(\mathcal{U}, T) = X_+ \cup_{X_0} X_-$. If all of T_-, T_+, and T_0 are nonempty then the induction hypothesis shows that all of X_-, X_+, and X_0 are contractible. In that case, $X(\mathcal{U}, T)$ is the union of contractible spaces along a contractible space, hence is contractible (since the inclusions are cofibrations). If one of the closed sets $T_- \subset \operatorname{Spec} P_-$ or $T_+ \subset \operatorname{Spec} P_+$ is empty then the other is not (since T is nonempty). In that case one of X_- or X_+ is empty and the other is not. Then $X(\mathcal{U}, T)$ is the union of a contractible set and an empty set, hence is contractible. The analysis of $X_+ \cap X_-$, below, rules out the possibility that $X_0 = \varnothing$ while both X_+ and X_- are nonempty.

Now we show that $X(\mathcal{U}, T) = X_+ \cup_{X_0} X_-$. First, if v is any valuation of P then either $v(\beta_1) \leq 0$, in which case v extends to P_-, or $v(\beta_1) \geq 0$. Thus, $\mathcal{V} \subset \mathcal{U}$ will intersect in T at (and extension of) v if and only if either $\mathcal{V}_-$ intersects in T_- at v or $\mathcal{V}_+$ intersects in T_+ at v.

It remains to show that $X_- \cap X_+ = X_0$. The containment $X_0 \subset X_- \cap X_+$ is clear. Suppose that $\mathcal{V} = \{i_1, \ldots, i_m\} \subset \mathcal{U}$, that $\mathcal{V}_-$ intersects in T_- at $v_- : P \to W_-$ with $v_-(\beta_1) \leq 0$, and that $\mathcal{V}_+$ intersects in T_+ at $v_+ : P \to W_+$ with $v_+(\beta_1) \geq 0$. If either of these inequalities is an equality then one of v_- or v_+ lies in T_0 and we are done. Otherwise, let

$$Q = \left(Q_{i_1} \oplus_P \cdots \oplus_P Q_{i_m}\right)^{\mathrm{sat}\sharp}.$$

Since v_- and v_+ both factor through Q, so does $(v_-, v_+) : P \to W_- \times W_+$. Let R be the saturated image of Q in $W_- \times W_+$.

Note that the map $\operatorname{Spec} R \to \operatorname{Spec} P$ carries the closed point to a specialization of the centers of v_- and v_+, hence into T. Since R has a valuation under which β_1 is strictly positive and a valuation under which it is strictly negative, neither the image β_1 nor the image of $-\beta_1$ in R^{gp} can lie in R. Therefore P has a sharp valuation v_0 factoring through R with $v_0(\beta_1) = 0$: use [Ogu18, Proposition I.2.4.1] to obtain a sharp valuation of the saturated image of R in R/Y where Y is the saturation of $\mathbf{Z}\beta_1$. Since this valuation is sharp, its center is the image of the closed point of $\operatorname{Spec} R$, hence lies in T. This gives a valuation of Q centered in T_0. (This rules out $T_0 = \varnothing$ when T_- and T_+ are both nonempty.) Therefore $\mathcal{V}_0$ meets in T_0 at v_0. We conclude that $X_- \cap X_+ = X_0$, as required. □

1.8 Cohomology of face-constructible sheaves

Let $\sigma = \operatorname{Cone} P$ for a monoid P. We write $\operatorname{Rhiz} P$ for the category of all monoid homomorphisms $P \to Q$, equipped with the birhizomic topology and $\operatorname{Spec} P$ for the small Zariski site of the spectrum of P.

There is a morphism of sites $\pi : \operatorname{Rhiz} P \to \operatorname{Spec} P$. A basic open subset of $\operatorname{Spec} P$ corresponds to a sharp localization $P \to \overline{P}$. On these basic open subsets, π^{-1} operates as the identity functor. It is easily verified that π^{-1} is left exact, since a pushout of sharp localizations is a sharp localization. Since basic open subsets have no nontrivial covers, it is equally immediate that π^{-1} preserves covers.

If $P \to Q$ is a finitely presented sharp localization, we call $\tau = \operatorname{Cone} Q$ a *face* of $\sigma = \operatorname{Cone} P$. In this case, there is an inclusion morphism $i : \operatorname{Spec} Q \to \operatorname{Spec} P$. The pullback functor i^{-1} on sheaves of abelian groups has a left adjoint $i_!$ (extension by zero for an open immersion). We have $i_!G(\rho) = G(\rho)$ if ρ is a face of τ, and $i_!G(\rho) = 0$ otherwise.

Lemma 1.8.1. *If F is a sheaf of abelian groups on $\sigma = \operatorname{Spec} P$ then F receives a surjection from a direct sum of sheaves $i_!G$, ranging over all face inclusions $i : \tau \to \sigma$ and G constant on τ.*

Proof. Let $i : \tau \to \sigma$ be the inclusion of a face of σ. We take G_τ to be the constant sheaf on τ with value $F(\tau)$. Then the tautological morphism $G_\tau \to i^{-1}F$ induces $i_!G_\tau \to F$. Then $\bigoplus_{i:\tau\to\sigma} i_!G_\tau \to F$ is surjective. □

We iterate the construction of Lemma 1.8.1 to obtain an exact sequence

$$\cdots F_{k+1} \to F_k \to F_{k-1} \to \cdots \to F \to 0$$

in which each F_k is a direct sum of sheaves of the form $j_!G$, where $j : \operatorname{Spec} \overline{P} \to \operatorname{Spec} P$ is the inclusion of a face and G is constant on $\operatorname{Spec} \overline{P}$.

Lemma 1.8.2. *Let $X \subseteq \operatorname{Spec} P$ be any open subset of $\operatorname{Spec} P$ where P is a finitely generated monoid. If F is a sheaf of abelian groups on X, the above resolution has $F_k = 0$ for $k > \dim X$.*

Proof. By induction on $\dim X$: if $\dim X = 0$, it is a finite disjoint union of points and Lemma 1.8.1 produces an isomorphism. Otherwise, note that the construction from Lemma 1.8.1 produces one nonzero summand for each maximal dimension face in the support of F, and that the value of that summand on that face is the same as the value of F there. Thus the kernel of $F_0 \to F$ is supported on a monoidal subspace of strictly smaller dimension. □

Lemma 1.8.3. *Let P be a finitely generated monoid and let $j : \operatorname{Spec} \overline{P} \to \operatorname{Spec} P$ be the inclusion of a face. Let G be a constant sheaf on $\operatorname{Spec} \overline{P}$. Then Theorem 1.2.1 holds for $F = j_!G$.*

Proof. Let F' be the constant sheaf on $\operatorname{Spec} P$ with the same value as G and let $F'' = F'/F$. Note that F'' is the pushforward of a constant sheaf from the closed complement of $\operatorname{Spec} \overline{P}$ (see [Sta20, Tag 02UT], for example).

We argue that all Čech cohomology groups of F' and of F'' vanish in positive degrees for sufficiently fine covers, and that $H^0(\operatorname{Rhiz} P, F') \to H^0(\operatorname{Rhiz} P, F'')$ is surjective. Since the formation of F'' commutes with restriction along any homomorphism $P \to P_1$, this will also imply the vanishing of the Čech cohomology of F' on every object of $\operatorname{Rhiz} P$. Since this reasoning applies to every finitely generated monoid P and all sufficiently fine birhizomic covers, and since the pushouts of finitely generated monoids are finitely generated, this will then imply the vanishing of the derived functor cohomology of F by the Čech-to-derived spectral sequence [Sta20, Tag 03F9].

The vanishing for F' is immediate because F' is constant and all $P \to Q_i$ in a birhizomic cover have a common intersection over the zero localization $P \to 0$, so the Čech cohomology coincides with the cohomology of a constant sheaf on a simplex. This also dispatches the case where $P \to \overline{P}$ is an isomorphism, since in that case $F' = F$.

Let K be the kernel of the sharp localization $P^{\mathrm{gp}} \to \overline{P}^{\mathrm{gp}}$. The prime ideal $P \setminus K$ is the complement of the image of j. Let $P_0 = P \cap K$. Then the prime ideals of P_0 are in bijection with the prime ideals of P containing $P \setminus K$, so we identify $\operatorname{Spec} P_0$ with the closed complement. We write $i : \operatorname{Spec} P_0 \to \operatorname{Spec} P$ for the inclusion and note that $F'' = i_* G'$ where G' is the constant sheaf on $\operatorname{Spec} P_0$. If $P \to Q_i$ is birhizomic and sharp then there is no open subset of $\operatorname{Spec} P$ that contains the image of $\operatorname{Spec} Q_i$ other than $\operatorname{Spec} P$ itself. In particular, the value of $\pi^{-1} i_* G'$ on Q_i is the constant value of G. By Lemma 1.4.3, every cover of $\operatorname{Rhiz} P$ can be refined by a family of sharp maps $P \to Q_i$, so we conclude that $H^0(\operatorname{Rhiz} P, F'')$ has the same constant value as does $H^0(\operatorname{Rhiz} P, F')$ and therefore that $H^0(\operatorname{Rhiz} P, F') \to H^0(\operatorname{Rhiz} P, F'')$ is surjective.

For the vanishing of the higher Čech cohomology groups of F'', we consider a birhizomic covering family $\mathcal{U} = \{P \to Q_i\}$ of P. The group $F''\big((Q_{i_1} \oplus_P \cdots \oplus_P Q_{i_n})^{\mathrm{sat}\sharp}\big)$ will vanish if $P \to (Q_{i_1} \oplus_P \cdots \oplus_P Q_{i_n})^{\mathrm{sat}\sharp}$ factors through $P \to \overline{P}$, and it will take the value G otherwise. Write $T \coloneqq \operatorname{Spec} P \setminus \operatorname{Spec} \overline{P}$ for the closed complement. Then the Čech complex of F'' on the cover $\mathcal{U}$ coincides with the normalized cochain complex of the constant sheaf G on the intersection complex $X(\{P \to Q_i\}, T)$ from Construction 1.7.3. By Lemma 1.4.3, we can replace $\{P \to Q_i\}$ with a refinement where every $P^{\mathrm{gp}} \to Q_i^{\mathrm{gp}}$ is bijective. Then cohomology vanishes by Proposition 1.7.4. □

Lemma 1.8.4. *Theorem 1.2.1 holds when P is finitely generated.*

Proof. Consider the finite resolution of Lemma 1.8.2 :

$$0 \to F_n \to \cdots \to F_1 \to F_0 \to F \to 0$$

Since P is finitely generated, each term in the resolution is a finite direct sum of sheaves $j_! G$, and Lemma 1.8.3 shows the sheaves $\pi^{-1} j_! G$ are acyclic on $\operatorname{Rhiz} P$. Therefore the hypercohomology spectral sequence degenerates at the E_2-page and the E_1-page is a single row:

$$0 \to H^0(\operatorname{Rhiz} P, \pi^{-1} F_n) \to \cdots \to H^0(\operatorname{Rhiz} P, \pi^{-1} F_0) \to 0 \tag{1.2}$$

Lemma 1.8.3 also implies that $H^0(\operatorname{Rhiz} P, \pi^{-1} F_q) = H^0(\operatorname{Spec} P, F_q)$. But the global sections functor on $\operatorname{Spec} P$ is exact because $\operatorname{Spec} P$ has no nontrivial covers. Therefore (1.2) is exact except in the last position, $H^i(\operatorname{Rhiz} P, \pi^{-1} F)$ vanishes for all $i > 0$, and $H^0(\operatorname{Rhiz} P, \pi^{-1} F) = H^0(\operatorname{Spec} P, F)$. □

Lemma 1.8.5. *If P is finitely generated monoid then the Čech cohomology $H^p(\mathcal{U}, F)$ vanishes for $p > 0$ and all sufficiently fine birhizomic covers $\mathcal{U}$ and all sheaves of abelian groups F on $\operatorname{Spec} P$.*

Proof. Let $\mathcal{U} = \{P \to Q_i\}$ be a birhizomic cover of P. Let $R = (Q_{i_0} \oplus_P \cdots \oplus_P Q_{i_s})^{\mathrm{sat}\sharp}$ be one of the monoids contributing to the Čech complex. The composition $\operatorname{Rhiz} R \to \operatorname{Rhiz} P \to \operatorname{Spec} P$ can also be factored as $\operatorname{Rhiz} R \to \operatorname{Spec} R \to \operatorname{Spec} P$. Therefore $\pi^{-1}(F)\big|_{\operatorname{Rhiz} R} = \pi^{-1}\big(F\big|_{\operatorname{Spec} R}\big)$. Furthermore, after refining the cover, we can assume that every Q_i, and therefore every R, is finitely generated relative to P. Since P is finitely generated, this implies that R is as well. Therefore, Lemma 1.8.4 applies to $\pi^{-1}(F)\big|_{\operatorname{Rhiz} R}$, and the higher cohomology of $\pi^{-1} F$ on all $\operatorname{Rhiz} R$ vanishes by Lemma 1.8.4. Therefore, the Čech spectral sequence for $\pi^{-1} F$ degenerates to the Čech complex at the E_1 page, so the Čech cohomology computes the cohomology of $\pi^{-1} F$ on $\operatorname{Rhiz} P$, hence vanishes in positive degrees. □

Proof of Theorem 1.2.1. We will argue by induction on p that the group $H^p(\operatorname{Rhiz} P, \pi^{-1} F)$ vanishes for $p > 0$. Suppose that $\alpha \in H^p(\operatorname{Rhiz} P, \pi^{-1} F)$. Choose a cover of $\operatorname{Rhiz} P$ by finitely many cones $P \to Q_i$ such that $\alpha\big|_{Q_i} = 0$ for all i. Induction implies that $H^q(\operatorname{Rhiz}\big((Q_{i_0} \oplus_P \cdots \oplus_P Q_{i_m})^\sharp\big), \pi^{-1} F) = 0$ for $0 < q < p$ and all $i_0, \ldots, i_m$. Therefore α can be represented by a Čech cohomology class with respect to the cover of $\operatorname{Rhiz} P$ by the $\operatorname{Rhiz} Q_i$, using the Čech-to-derived spectral sequence and inductive vanishing.

Using Proposition 1.6.4, we can find a family of finitely presented homomorphisms $Q_i \to W_i$ such that $P \to Q_i$ is birhizomic at $Q_i \to W_i$ and the maps $|\operatorname{Cone}(W_i)|$ cover $|\operatorname{Cone}(P)|$. Reindexing I may be necessary to factor Q_i through multiple W_i's. By Proposition 1.6.3, each of the $|\operatorname{Cone}(W_i)|$ has constructible image in

$|\mathrm{Cone}(P)|$, and therefore Proposition 1.6.2 shows that a finite subset of them cover $|\mathrm{Cone}(P)|$. We replace the Q_i by this finite collection.

Since there are only finitely many Q_i, and since each is finitely presented relative to P, there is a finitely generated monoid P' and homomorphism $P' \to P$ such that each Q_i is induced by base change from a finitely generated extension $P' \to Q_i'$. By Corollary 1.5.6, it is possible, for each i, to choose P' and the Q_i' so that each $P' \to Q_i'$ is birhizomic at $Q_i' \to W_i$. Since there are only finitely many Q_i, a single P' works for all i. The birhizomic locus of $P' \to Q_i'$ is a union of constructible subsets of $|\mathrm{Cone}(Q_i')|$ by Proposition 1.6.4. A finite collection of these suffice to cover the images of $|\mathrm{Cone}(W_i)|$ by Proposition 1.6.2. Relabeling indices if necessary, there is therefore a family of homomorphisms $P' \to Q_i' \to W_i'$ such that $P' \to Q_i'$ is birhizomic at $Q_i' \to W_i'$ and, if W_i'' denotes the extension of W_i' to P, then the $|\mathrm{Cone}(W_i'')|$ cover $|\mathrm{Cone}(P)|$.

For each finitely presented monoid homomorphism $P' \to P'_\alpha$ factoring $P' \to P$, let $W'_{\alpha i}$ be the pushout of W_i' along $P' \to P'_\alpha$. In the filtered colimit, the $|\mathrm{Cone}\left(\mathrm{colim}_\alpha W'_{\alpha i}\right)|$ cover $|\mathrm{Cone}(P)|$, so Proposition 1.6.6 implies that there is some index α where the $|\mathrm{Cone}(W'_{\alpha i})|$ already cover $|\mathrm{Cone}(P'_\alpha)|$. Writing $Q'_{\alpha i}$ for the pushout of Q_i', note that $P'_\alpha \to Q'_{\alpha i}$ is birhizomic at $Q'_{\alpha i} \to W'_{\alpha i}$ by Lemma 1.3.8. Therefore we can replace P' by P'_α and assume that the $|\mathrm{Cone}(W_i')|$ cover $|\mathrm{Cone}(P')|$.

For each tuple $\mathbf{i} = (i_0, \ldots, i_s)$, let $Q_{\mathbf{i}} = (Q_{i_0} \oplus_P \cdots \oplus_P Q_{i_s})^\sharp$. For each $\mathbf{i}$, let $P_{\mathbf{i}}$ be the sharpening of $P \to Q_{\mathbf{i}}$. The homomorphism $P \to P_{\mathbf{i}}$ is itself finitely presented by Lemma 1.3.3. After enlarging P', we assume that each $P \to P_{\mathbf{i}}$ is the extension to P of a sharp localization $P' \to P'_{\mathbf{i}}$. Since $P_{\mathbf{i}}$ only depends on the distinct indices in $\mathbf{i}$, and since there are only finitely many Q_i, there are only finitely many distinct $P_{\mathbf{i}}$, and therefore we can find a single enlargement P' such that every $P_{\mathbf{i}}$ is the extension of the sharp localization $P'_{\mathbf{i}}$. Now we take F' to be the pushforward of F along the morphism $\operatorname{Spec} P \to \operatorname{Spec} P'$. Let us write $Q'_{\mathbf{i}} = (Q'_{i_0} \oplus_P \cdots \oplus_P Q_{i_s})^\sharp$. Then by definition of π^{-1}, the value $\pi^{-1}F'(Q'_{\mathbf{i}})$ is the value of F' on the minimal face $\operatorname{Spec} P'_{\mathbf{i}}$ through which $\operatorname{Spec} Q_{\mathbf{i}} \to \operatorname{Spec} P'$ factors. Since $\operatorname{Spec} P'_{\mathbf{i}}$ pulls back to $\operatorname{Spec} P_{\mathbf{i}}$ by construction, we have $\pi^{-1}F'(Q'_{\mathbf{i}}) = \pi^{-1}F(Q_{\mathbf{i}})$.

Our arrangements ensure that the Čech complex of $\pi^{-1}F'$ with respect to the cover $\{P' \to Q_i'\}$ of $\mathrm{Rhiz}(P')$ is identical to the Čech complex of $\pi^{-1}F$ with respect to the cover $\{Q_i\}$ of $\mathrm{Rhiz}(P)$. But the Čech cohomology of $\pi^{-1}F'$ on $\mathrm{Rhiz}(P')$ vanishes in positive degrees by Lemma 1.8.5. Since F' is the pushforward of F, we also have

$$H^0(\{P \to Q_i\}, \pi^{-1}F) = H^0(\{P' \to Q_i'\}, \pi^{-1}F') = H^0(\mathrm{Rhiz}(P'), \pi^{-1}F') = H^0(\mathrm{Rhiz}(P), \pi^{-1}F)$$

and we may conclude. □

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